

A THESIS TYPESET WITH TYPST

The Faculty of Sciences
University of ABC for
the doctoral degree

Doctor rerum typorum (Dr. rer. typ.)

submitted by

OTTO MATIC

born in

HACHTEL

To XYZ
A quotation

ACKNOWLEDGEMENTS

I thank coffee and my friends. In that order.

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1.1 FIGURES

In this template, we have figures, as shown in Fig. 1.

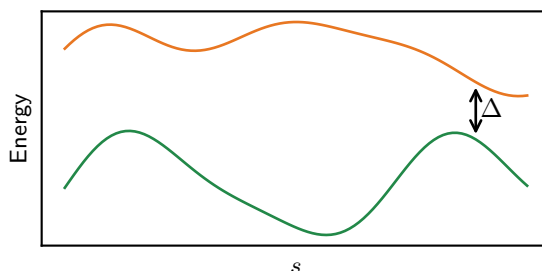


Fig. 1: An example figure. The caption font is slightly smaller than the body font.

1.2 CITATIONS

We also have nifty citation tools, for instance `@mike_and_ike[!]` will render as a prose citation, Nielsen and Chuang [1]. We can also add specifications, with `@mike_and_ike[!: p. 100]`, rendering as Nielsen and Chuang [1, p. 100]. The normal citation `@mike_and_ike[p. 100]` works as well: [1, p. 100]. Citations use two custom styles defined in `csl/`. The prose citation uses `short_and_sweet.csl`, while the bibliography section uses `thoughtful_and_coolheaded.csl` plus some show rules.

1.3 REFABLE-BOXES

In order to implement theorems and such, there's a higher-level function called `refable-box` (in the sense that it can be referenced). For example the theorem is implemented as

```
#let theorem(body) = refable-box(body, kind: "Theorem")
```

which gives

Theorem 1

Peach-flavored iced tea is superior to lemon-flavored iced tea.

Theorems also usually are followed by proofs, which can be called right away with `#proof(body)`

Proof: It's obvious.

1.4 ACRONYMS

Acronyms can be defined in the `acrodictionary` array at `inputs.typ`. Then, the template will do its best to identify them all, turn any later use of the full expression into an acronym, and add a link to a table at the end. The first use of the expression will have the acronym in parentheses afterwards. The main caveat is that the first mention in the text must be the full expression, the script will not replace an acronym with the expression, because I'm too lazy to manage capitalization edge cases. For example, super useful acronym (SUA).

- Super useful acronym

1.5 ALGORITHMS

Quantum mechanics is not going to be replaced by something that's easier.

— Carlton Caves

In June 1925, Heisenberg fled the Göttingen pollen and beached Helgoland, a collection of orange rocks on the North Sea. He was very productive there, and, in the resulting publication [2], he stated that the product of some physical values, when using the calculation rules of the contemporary theory of quanta, is not commutative — this is recognized by many as the birth of modern quantum mechanics.

In June 2025, a centennial conference was held on the same island to celebrate the result and ponder its consequences. It illuminated the amount of work built on top of this achievement, and simultaneously, the depth of the still unknown. Technological advancement has been remarkable: arrays of atoms can be precisely controlled [3,4]; interactions between gravity and quantum physics are a future milestone rather than a far-off dream [5], and, of special interest to this thesis, quantum computers are, though today not significantly more powerful than classical computers in many regards, a reality [1].

2.1 QUANTUM COMPUTERS

These quantum devices, as foretold in a slightly overused quotation of Richard Feynman, would be used to simulate the dynamical evolution of quantum systems, because “nature isn’t classical”, unfortunately [6]. It has not however been proven that quantum computers can perform computations that classical computers cannot perform [7], but it is widely believed that quantum dynamics cannot, in general, be efficiently simulated on classical computers, and the conjecture is used as a basis for claims of theoretical quantum advantage [8,9]. In fact, if one could efficiently simulate quantum dynamics on a classical device such as a laptop or a supercomputer, then any quantum algorithm could be efficiently run on the device by simulating the quantum computer that runs the program.

If that is not the case, quantum computers have a good chance of exceeding classical capabilities. For that to happen, we must design new quantum algorithms that conform to the constraints of quantum devices, and this requires us to radically change our way of thinking about algorithms. In short, the quantumness of quantum computers must be fully leveraged and understood for them to surpass classical computers: a plane can drive on a road just like a car, but it’s more useful when it flies.

2.2 THE TORIC CODE

Toric code are essentially the definition of logical qubits on a periodic lattice (a torus) of physical qubits through the topological properties arising from the definition of two special operators. Kitaev [10] introduced the concept. Let us define the plaquette and vertex operators.

$$\begin{aligned} A_p &= \sum_{i \in \vdash_p} Z_i \\ B_k &= \sum_{i \in \square_k} X_k \end{aligned} \tag{1}$$

where $+_p$ is the set of operators acting on the edges around a vertex v_p (the four ends of the cross), whereas $\square_k$ contains operators centered on a face f_k , acting on its four edges. These two operators always commute, since the number of overlapping vertices is always even.

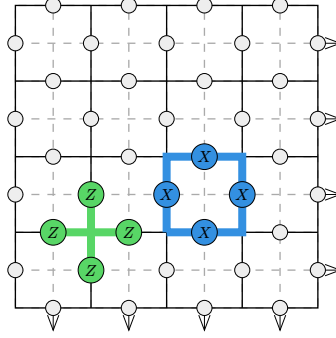


Fig. 2: The spins are on the edges of a dual lattice, the blue face represents the B operator, and the green cross represents the A operator, and the arrows mean the circle is the periodic boundary, i.e. a circle with an arrow represents a circle on the other side of the lattice

Consider the Hamiltonian;

$$H = - \sum_p A_p - \sum_k B_k \quad (2)$$

It has four distinct degenerate ground states: the entirely spin 0 state S_0 , also known as the “loop vacuum”, which includes any loop that doesn’t wind around the boundary, since it can be deformed using B back to the vacuum.

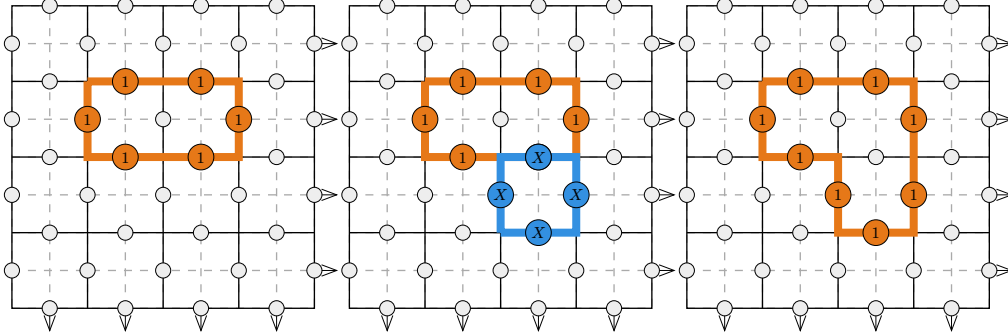


Fig. 3: We can deform any non winding path into the empty lattice, or into any other non winding path, using the B operators, which “flips” a face-full of spins

The three other group are the vertically winding string, S_v , the horizontal string, S_h , and their sum S_S . Each group is linearly independent with respect to the A and B operators.

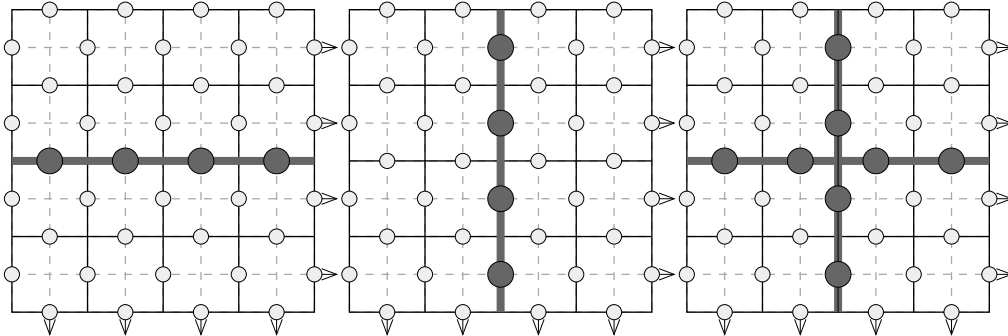


Fig. 4: Three of the four independent degenerate ground state groups. The fourth is the empty lattice.

These four ground states are generated by winding Pauli strings.

If we identify each group with a state,

$$\begin{aligned} S_0 &= |00\rangle \\ S_h &= |10\rangle \\ S_v &= |01\rangle \\ S_s &= |11\rangle. \end{aligned} \tag{3}$$

we may switch from a two-qubit state to another, by applying a vertical string, either X_v or Z_v , or a horizontal string, either X_h or Z_h .

$$\begin{aligned} X_h|00\rangle &= |10\rangle = \tilde{X}_1|00\rangle \\ Z_v|10\rangle &= -|10\rangle = \tilde{Z}_1|10\rangle \\ X_v|00\rangle &= |01\rangle = \tilde{X}_2|00\rangle \\ Z_h|01\rangle &= -|01\rangle = \tilde{Z}_2|01\rangle \end{aligned} \tag{4}$$

We then have two qubits encoded in this lattice, which are protected against any stabilizer operation. More formally, there exists a protected subspace $\mathcal{L}$ of dimension 4:

$$\mathcal{V} := \{|\xi\rangle : A_p|\xi\rangle = |\xi\rangle, B_k|\xi\rangle = |\xi\rangle, \forall p, k\} \tag{5}$$

In this protected subspace, any undesired A_p or B_k applied to the system leaves it invariant. We may, however, create abelian anyonic excitations with string operators S^z and S^x , which would break the invariance:

$$\begin{aligned} S^z(t) &= \sum_{j \in t} Z_j \\ S^x(t') &= \sum_{j \in t'} X_j \end{aligned} \tag{6}$$

where t and t' are non-closed paths (“strings”) on the lattice. Two particles are created, one at each end of the string. Z particles are called *electric charges* and X particles *magnetic vortices*.

2.3 OUTLOOK

We can only see a short distance ahead, but we can see plenty there that needs to be done.

— Alan Turing

Feynman predicted that computers might become quantum by 2050 [11]. Though he could not predict the difficulties we would meet along the way, it seems that his idea of quantum mechanical computers is slowly turning out to be true, as our NISQ computers turn fault-tolerant, and the algorithms that yesterday were beyond our grasp are run on today's devices. Looking forward, we hope that quantum digital cooling for fermionic systems beyond toy models becomes attainable, if not commonplace, on tomorrow's devices.

On a more philosophical note, it is startling to see how much of quantum mechanics is still a mystery, and the disagreements on what it tells us about reality are witnesses to its enduring strangeness. Quantum computing helps us lift the veil of this unknown, using a mystery to solve itself; and still, it matters little how advanced quantum devices get: our eyes only see the classical world, and only through it do we get closer to this stranger truth. On Helgoland, Anton Zeilinger said that quantum mechanics links classical data with classical data. To reformulate this in Feynman's words, "nature is classical, dammit!"

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APPENDIX

You can also write stuff after you're done with the main text if that's your thing.

A AND THE NUMBERING WILL CHANGE

A.1 AS EXPECTED OF THE APPENDIX

A.2 AND SO ON

A.2.1 AND SO FORTH

B IT SHOULD FOLLOW THE RULES

B.1 AS EXPECTED

ACRONYMS

- **ANAC** another acronym
- **SUA** super useful acronym

COLOPHON

This thesis was written in Typst rather than the usual LaTeX, on a template built *ab initio*. The template was nonetheless strongly inspired by the venerable ClassicThesis LaTeX package.

I used the fonts

- Arial for the cover,
- Tex Gyre Heros for titles,
- Gentium Book Basic for body text,
- New Computer Modern Math for math,
- Roboto Mono for monospace, and
- AuriocusKalligraphicus for chapter numbers and some diagrams.

This document was last compiled on Monday 17/08/2026, with Typst 0.15.1.